RISE OF ROBOT RADIOLOGISTS

1.INTRODUCTION

Artificial intelligence algorithms, particularly deep learning, have demonstrated remarkable progress in image-recognition tasks.

In radiology practice, trained physicians visually assessed medical images for the detection, characterization and monitoring of diseases. AI methods excel at automatically recognizing complex patterns in imaging data and providing quantitative, rather than qualitative, assessments of radiographic characteristics.

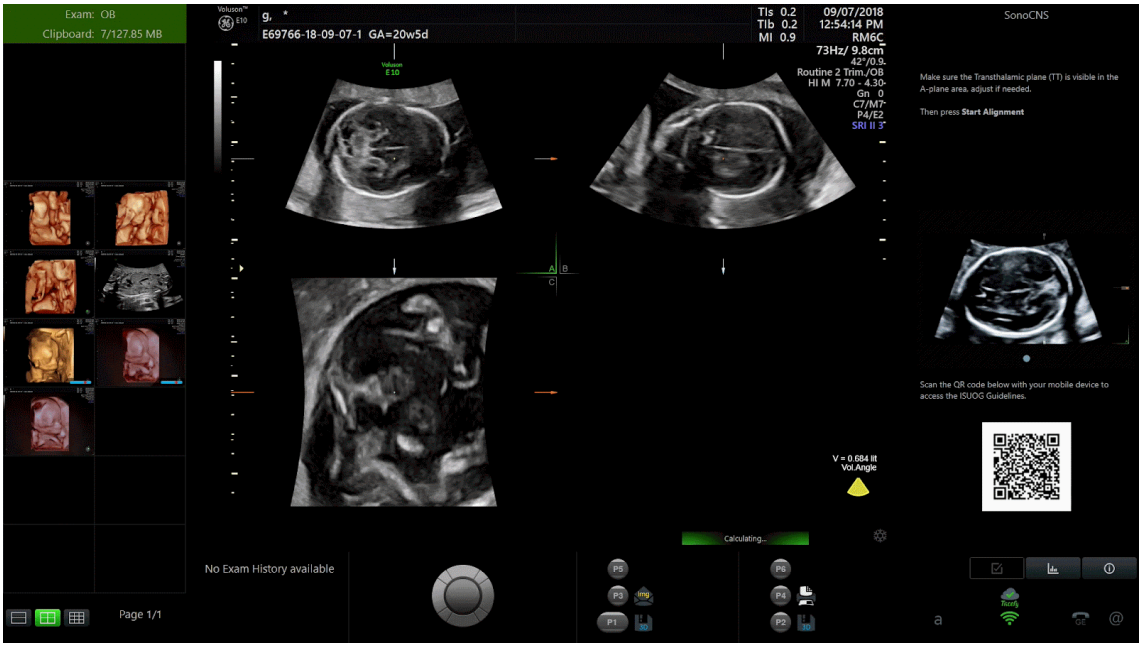

Fig. 1.1 AI softwares detecting abnormalities in body(GE Healthcare's SonoCNS, an artificial intelligence tool that automates fetal brain assessment and measurements)

Artificial intelligence (AI) has recently made substantial strides in perception (the interpretation of sensory information), allowing machines to better represent and interpret complex data. This has led to major advances in applications ranging from web search and self-driving vehicles to natural language processing and computer vision — tasks that until a few years ago could be done only by humans.

Within health care, AI is becoming a major constituent of many applications, including drug discovery, remote patient monitoring, medical diagnostics and imaging, risk management, wearables, virtual assistants and hospital management.

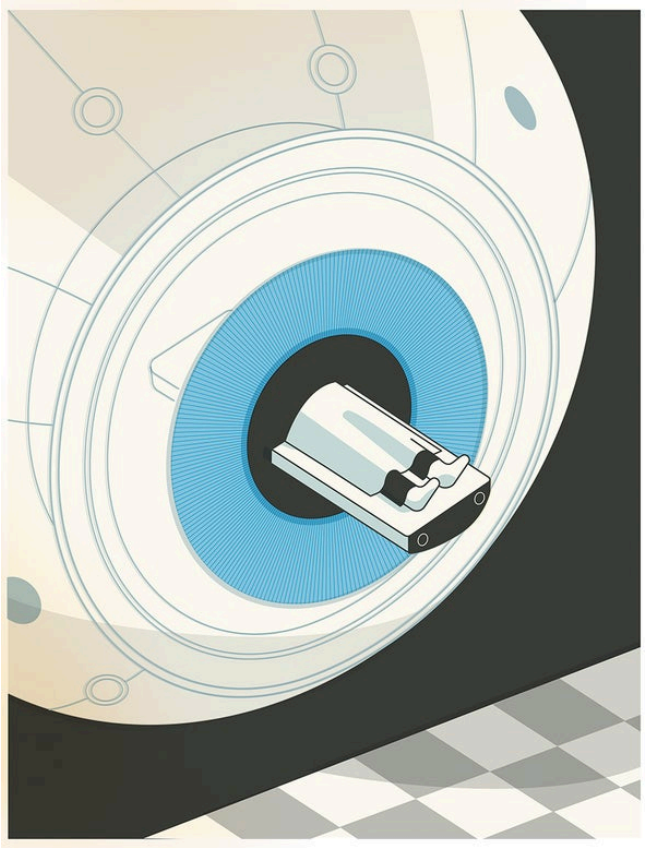

Fig. 1.2 Futuristic Image processing units in future

Medical fields that rely on imaging data, including radiology, pathology, dermatology and ophthalmology, have already begun to benefit from the implementation of AI methods . Within radiology, trained physicians visually assess medical images and report findings to detect, characterize and monitor diseases. Such assessment is often based on education and experience and can be, at times, subjective. In contrast to such qualitative reasoning, AI excels at recognizing complex patterns in imaging data and can provide a quantitative assessment in an automated fashion. More accurate and reproducible radiology assessments can then be made when AI is integrated into the clinical workflow as a tool to assist physicians.

2.WHAT IS AI AND DEEP LEARNING?

Artificial intelligence (AI) is a wide-ranging branch of computer science concerned with building smart machines capable of performing tasks that typically require human intelligence. AI is an interdisciplinary science with multiple approaches, but advancements in machine learning and deep learning are creating a paradigm shift in virtually every sector of the tech industry.

Deep learning is a subset of machine learning that is based on a neural network structure loosely inspired by the human brain. Such structures learn discriminative features from data automatically, giving them the ability to approximate very complex nonlinear relationships.

Deep Learning methods are a modern update to Artificial Neural Networks that exploit abundant cheap computation.

They are concerned with building much larger and more complex neural networks and, as commented on above, many methods are concerned with very large datasets of labelled analog data, such as image, text. audio, and video. Accordingly, the deep learning algorithm gets a lot of attention these days to solve various problems in medical imaging fields. One example is to detect disease or abnormalities from X-ray images and classify them into several disease types or severities in radiology . This kind of task has been executed based on the various ML algorithms with proper optimization, theoretical or empirical approaches.

2.1 CONVOLUTIONAL NEURAL NETWORK (CNN)

CONVOLUTIONAL LAYER

The convolutional layer is the core building block of a CNN. The layer's parameters consist of a set of learnable filters (or kernels), which have a small receptive field, but extend through the full depth of the input volume. During the forward pass, each filter is convolved across the width and height of the input volume, computing the dot product between the entries of the filter and the input and producing a 2-dimensional activation map of that filter. As a result, the network learns filters that activate when it detects some specific type of feature at some spatial position in the input.

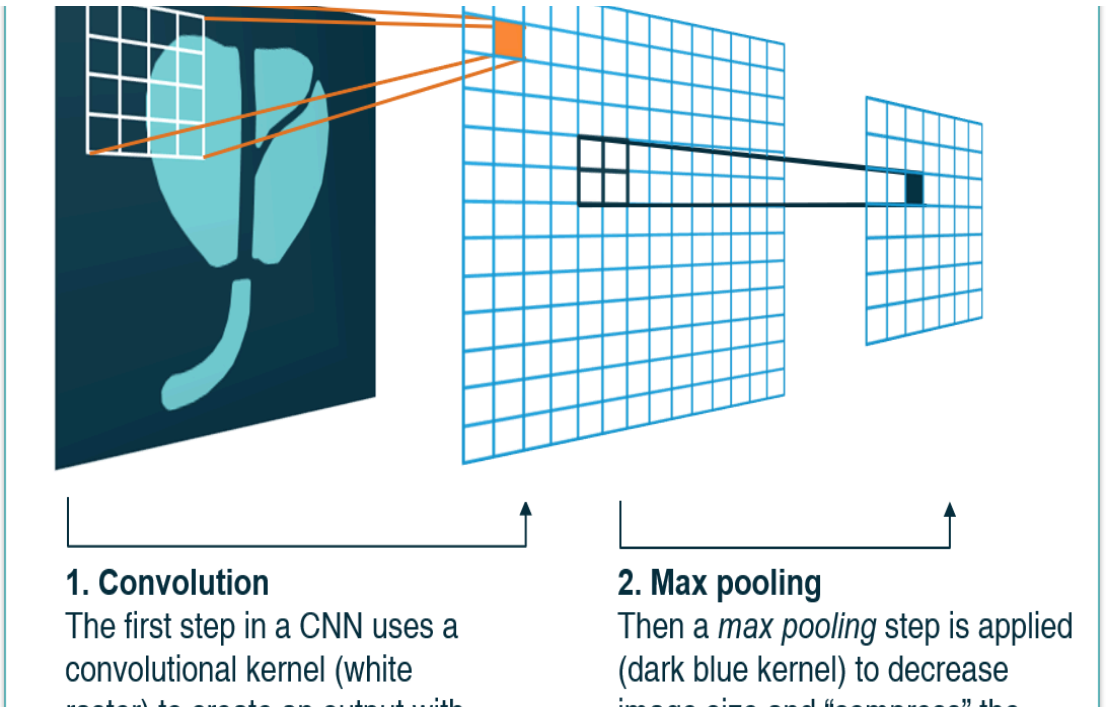

Fig. 2.1 Basic convolution pooling and processing

Stacking the activation maps for all filters along the depth dimension forms the full output volume of the convolution layer. Every entry in the output volume can thus also be interpreted as an output of a neuron that looks at a small region in the input and shares parameters with neurons in the same activation map.

2.2 TYPICAL CNN ARCHITECTURE

When dealing with high-dimensional inputs such as images, it is impractical to connect neurons to all neurons in the previous volume because such a network architecture does not take the spatial structure of the data into account. Convolutional networks exploit spatially local correlation by enforcing a sparse local connectivity pattern between neurons of adjacent layers: each neuron is connected to only a small region of the input volume.

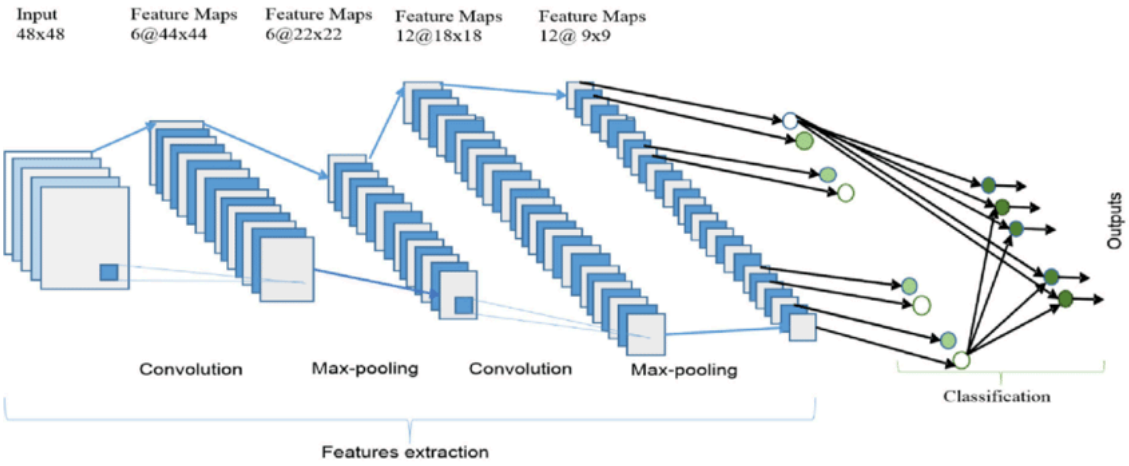

Fig. 2.2 Image processing done via convolution technology

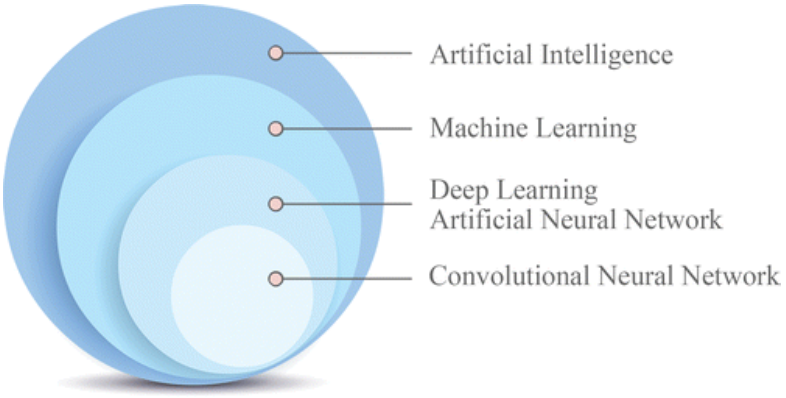

Fig. 2.3 Classifications of AI

The extent of this connectivity is a hyperparameter called the receptive field of the neuron. The connections are local in space (along width and height), but always extend along the entire depth of the input volume. Such an architecture ensures that the learnt filters produce the strongest response to a spatially local input pattern.

3.WHAT IS RADIOLOGY?

Radiology is the medical discipline that uses medical imaging to diagnose and treat diseases within the bodies of animals, including humans.

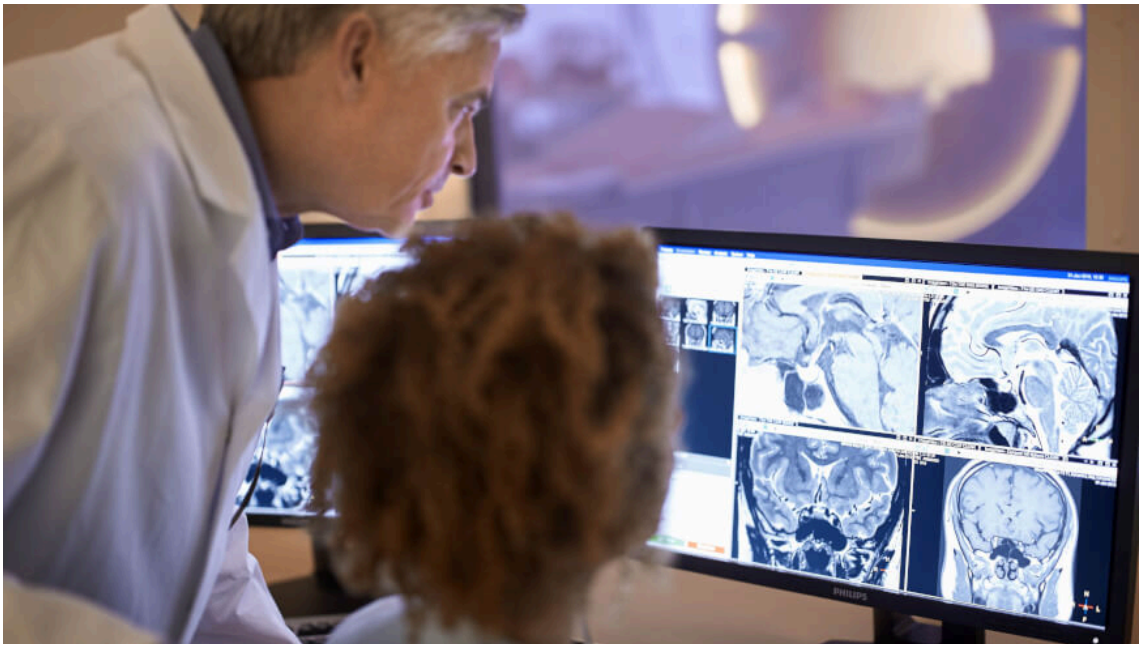

Fig. 3.1 Radiologists and doctors examining an MRI

A variety of imaging techniques such as X-ray radiography, ultrasound, computed tomography (CT), nuclear medicine including positron emission tomography (PET), fluoroscopy, and magnetic resonance imaging (MRI) are used to diagnose or treat diseases. Interventional radiology is the performance of usually minimally invasive medical procedures with the guidance of imaging technologies such as those mentioned above.

4.WHAT DOES A RADIOLOGIST DO?

Radiologists highly specialized medical doctors. Diagnostic radiology uses high-tech imaging (X-rays, ultrasound, CT scanning, MRI, nuclear medicine etc.) to

visualize everything from bone breaks to cancer. Interventional radiology involves minimally-invasive techniques for treatments that used to require surgery.

Fig. 4.1 Showing why radiologists get less attentions

If a radiologist misses that tiny, crucial detail on an image there could be a delay in treatment, unnecessary testing or over-treatment. "Behind the scenes" your radiologist is playing a key role in today's diagnosis, treatment planning and cost effective care.

Imaging technology, complexity and capability is rapidly expanding; even becoming a major tool in research trials. Patients may be receiving care from multiple providers or institutions and accurate and timely electronic image and report sharing is vital.

5.ARTIFICIAL INTELLIGENCE METHODS IN MEDICAL IMAGING

Machine learning algorithms based on predefined engineered features

Traditional artificial intelligence (AI) methods rely largely on predefined engineered feature algorithms with explicit parameters based on expert knowledge. Such features are designed to quantify specific radiographic characteristics, such as the 3D shape of a tumour or the intratumoral texture and distribution of pixel intensities (histogram). A subsequent selection step ensures that only the most relevant features are used. Statistical machine learning models are then fit to these data to identify potential imaging-based biomarkers. Examples of these models include support vector machines and random forests.

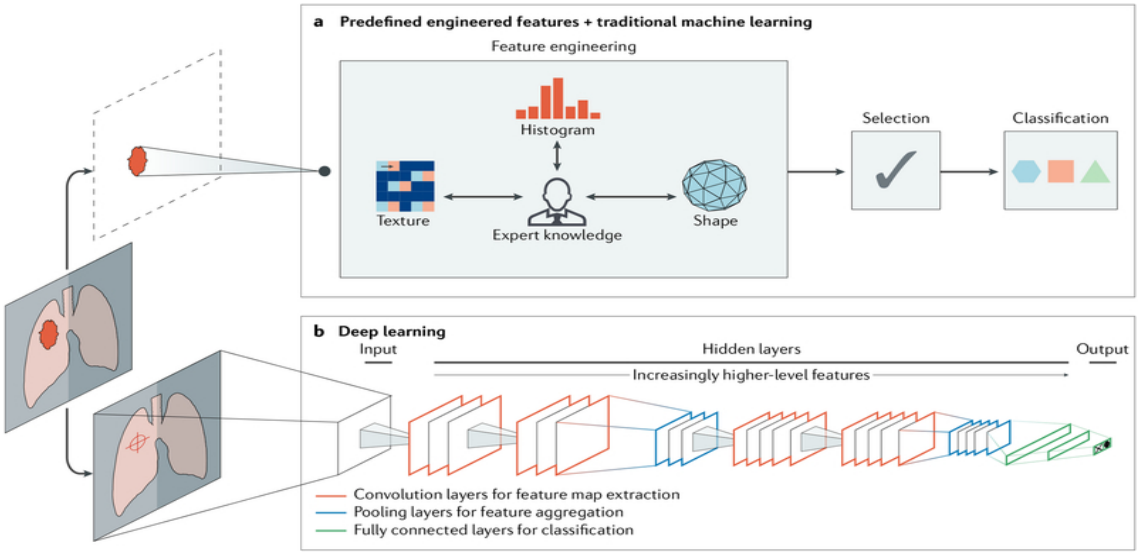

Fig. 5.1 Showing two ways which we use AI in medical imaging

This schematic outlines two artificial intelligence (AI) methods for a representative classification task, such as the diagnosis of a suspicious object as either

benign or malignant. a | The first method relies on engineered features extracted from regions of interest on the basis of expert knowledge. Examples of these features in cancer characterization include tumour volume, shape, texture, intensity and location. The most robust features are selected and fed into machine learning classifiers. b | The second method uses deep learning and does not require region annotation — rather, localization is usually sufficient. It comprises several layers where feature extraction, selection and ultimate classification are performed simultaneously during training. As layers learn increasingly higher-level features (Box 1), earlier layers might learn abstract shapes such as lines and shadows, while other deeper layers might learn entire organs or objects. Both methods fall under radiomics, the data-centric, radiology-based research field.

5.1 DEEP LEARNING ALGORITHMS

Recent advances in AI research have given rise to new, non-deterministic, deep learning algorithms that do not require explicit feature definition, representing a fundamentally different paradigm in machine learning. The underlying methods of deep learning have existed for decades. However, only in recent years have sufficient data and computational power become available. Without explicit feature predefinition or selection, these algorithms learn directly by navigating the data space, giving them superior problem-solving capabilities. While various deep learning architectures have been explored to address different tasks, convolutional neural networks (CNNs) are the most prevalent deep learning architecture typologies in medical imaging today. A

typical CNN comprises a series of layers that successively map image inputs to desired end points while learning increasingly higher-level imaging features . Starting from an input image, 'hidden layers' within CNNs usually include a series of convolution and pooling operations extracting feature maps and performing feature aggregation, respectively. These hidden layers are then followed by fully connected layers providing high-level reasoning before an output layer produces predictions. CNNs are often trained end-to-end with labelled data for supervised learning. Other architectures, such as deep autoencoders and generative adversarial networks, are more suited for unsupervised learning tasks on unlabelled data. Transfer learning, or using pre-trained networks on other data sets, is often utilized when dealing with scarce data.

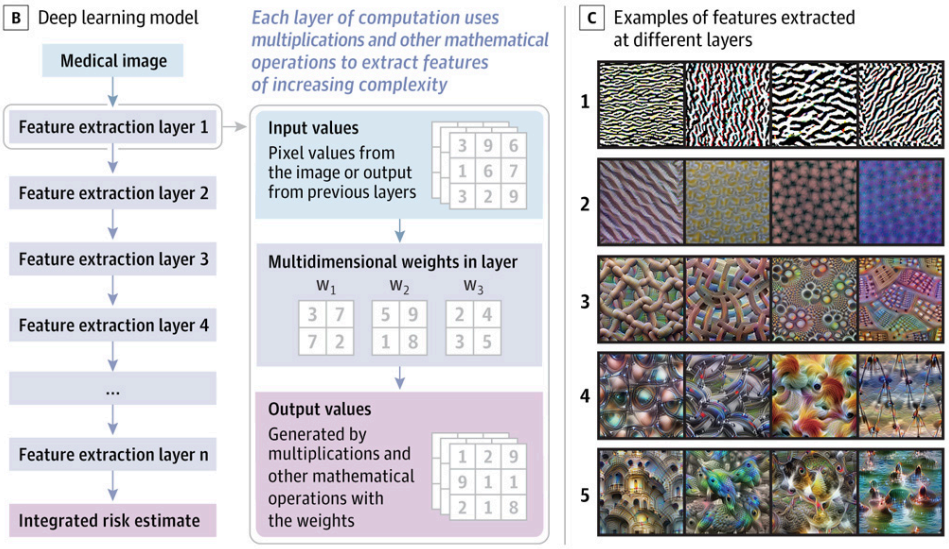

Fig. 5.2 Showing deep learning model

Within health care, AI is becoming a major constituent of many applications, including drug discovery, remote patient monitoring, medical diagnostics and imaging, risk management, wearables, virtual assistants and hospital management. Many domains

with big data components such as the analysis of DNA and RNA sequencing data are also expected to benefit from the use of AI. Medical fields that rely on imaging data, including radiology, pathology, dermatology and ophthalmology, have already begun to benefit from the implementation of AI methods. Within radiology, trained physicians visually assess medical images and report findings to detect, characterize and monitor diseases. Such assessment is often based on education and experience and can be, at times, subjective. In contrast to such qualitative reasoning, AI excels at recognizing complex patterns in imaging data and can provide a quantitative assessment in an automated fashion. More accurate and reproducible radiology assessments can then be made when AI is integrated into the clinical workflow as a tool to assist physicians.

Since CNN architecture is composed of many layers, a number of parameters for training can reach millions. This means a lot of data is needed for training to acquire competent accuracy. The number of data depends on task purpose and image characteristics. For instance, at least 1,000 images per class are necessary to get a competent result in a classification task if one trains the data from scratch. However, data collection is usually very difficult and even more hard if one also needs labeled data. To overcome this problem, data augmentation is also tried in general task which generates images from a limited number of data using image transformation methods such as rotation, translation, scaling, flipping, so on.

6.EXAMPLES OF CLINICAL APPLICATION AREAS OF ARTIFICIAL INTELLIGENCE IN ONCOLOGY

6.1 RADIOLOGY-BASED

6.1.1 Thoracic imaging

Lung cancer is one of the most common and deadly tumours. Lung cancer screening can help identify pulmonary nodules, with early detection being lifesaving in many patients. Artificial intelligence (AI) can help in automatically identifying these nodules and categorizing them as benign or malignant.

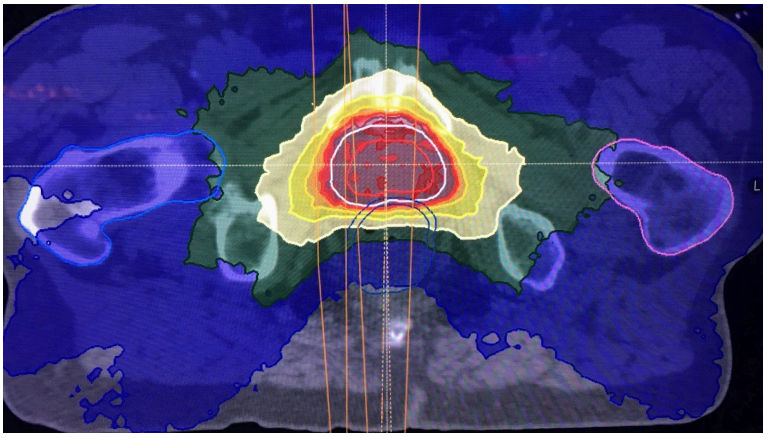

Fig. 6.1 AI showing cancer part in red

6.1.2 Abdominal and pelvic imaging

With the rapid growth in medical imaging, especially computed tomography (CT) and magnetic resonance imaging (MRI), more incidental findings, including liver lesions, are identified. AI may aid in characterizing these lesions as benign or malignant and prioritizing follow-up evaluation for patients with these lesions.

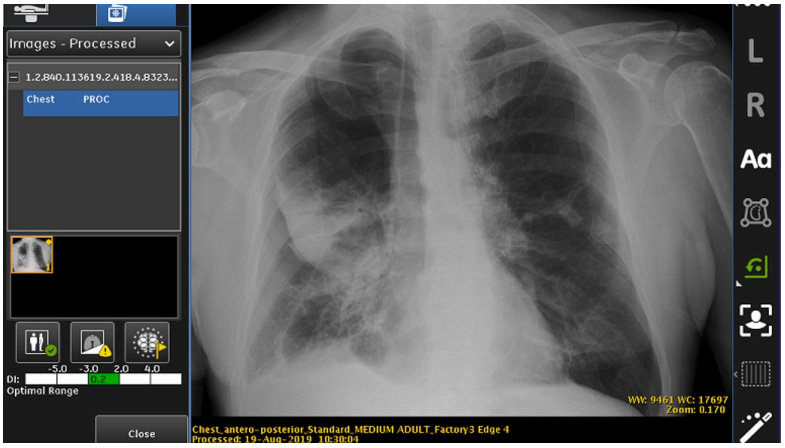

Fig. 6.2 AI tool being used to search abnormalities

6.1.3 Colonoscopy

Colonic polyps that are undetected or misclassified pose a potential risk of colorectal cancer. Although most polyps are initially benign, they can become malignant over time. Hence, early detection and consistent monitoring with robust AI-based tools are critical.

6.1.4 Mammography.

Screening mammography is technically challenging to expertly interpret. AI can assist in the interpretation, in part by identifying and characterizing microcalcifications (small deposits of calcium in the breast).

6.1.5 Brain imaging.

Brain tumours are characterized by abnormal growth of tissue and can be benign, malignant, primary or metastatic; AI could be used to make diagnostic predictions.

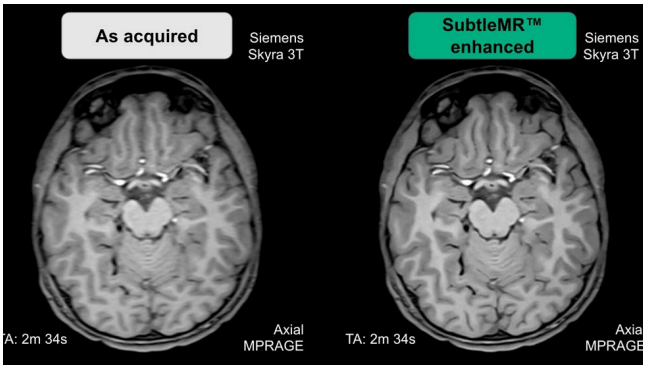

Fig. 6.3 AI showing enhanced images of same picture

6.1.6 Radiation oncology.

Radiation treatment planning can be automated by segmenting tumours for radiation dose optimization. Furthermore, assessing response to treatment by monitoring over time is essential for evaluating the success of radiation therapy efforts. AI is able to perform these assessments, thereby improving accuracy and speed.

6.2 NON-RADIOLOGY-BASED

6.2.1 Dermatology

Diagnosing skin cancer requires trained dermatologists to visually inspect suspicious areas. With the large variability in sizes, shades and textures, skin lesions are rather challenging to interpret. The massive learning capacity of deep learning algorithms qualifies them to handle such variance and detect characteristics well beyond those considered by humans.

6.2.2 Pathology

The quantification of digital whole-slide images of biopsy samples is vital in the accurate diagnosis of many types of cancers. With the large variation in imaging

hardware, slide preparation, magnification and staining techniques, traditional AI methods often require considerable tuning to address this problem. More robust AI is able to more accurately perform mitosis detection, segment histologic primitives (such as nuclei, tubules and epithelium), count events and characterize and classify tissue.

6.2.3 DNA and RNA sequencing.

The ever-increasing amount of available sequencing data continues to provide opportunities for utilizing genomic end points in cancer diagnosis and care. AI-based tools are able to identify and extract high-level features correlating somatic point mutations and cancer types as well as predict the effect of mutations on sequence specificities of RNA-binding and DNA-binding proteins

7.AI IN MEDICAL IMAGING

The primary driver behind the emergence of AI in medical imaging has been the desire for greater efficacy and efficiency in clinical care. Radiological imaging data continues to grow at a disproportionate rate when compared with the number of available trained readers, and the decline in imaging reimbursements has forced health-care providers to compensate by increasing productivity. These factors have contributed to a dramatic increase in radiologists' workloads. Studies report that, in some cases, an average radiologist must interpret one image every 3–4 seconds in an 8-hour workday to meet workload demands. As radiology involves visual perception as well as decision making under uncertainty, errors are inevitable — especially under such constrained conditions.

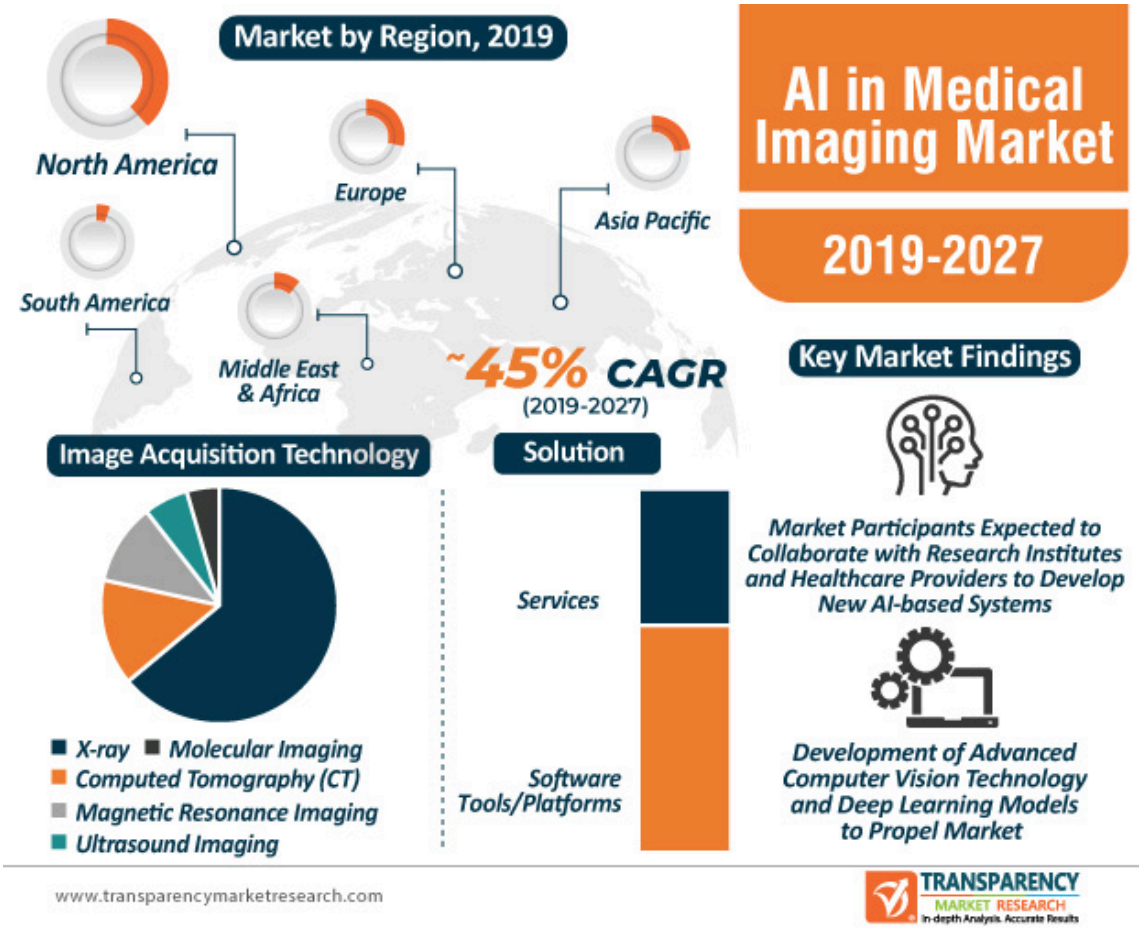

Fig. 7.1 Study by TMR markets of AI medical Imaging around the world

A seamlessly integrated AI component within the imaging workflow would increase efficiency, reduce errors and achieve objectives with minimal manual input by providing trained radiologists with pre-screened images and identified features. Therefore, substantial efforts and policies are being put forward to facilitate technological advances related to AI in medical imaging. Almost all image-based radiology tasks are contingent upon the quantification and assessment of radiographic characteristics from images. These characteristics can be important for the clinical task at hand, that is, for the detection, characterization or monitoring of diseases. The

application of logic and statistical pattern recognition to problems in medicine has been proposed since the early 1960s. As computers became more prevalent in the 1980s, the AI-powered automation of many clinical tasks has shifted radiology from a perceptual subjective craft to a quantitatively computable domain. The rate at which AI is evolving radiology is parallel to that in other application areas and is proportional to the rapid growth of data and computational power.

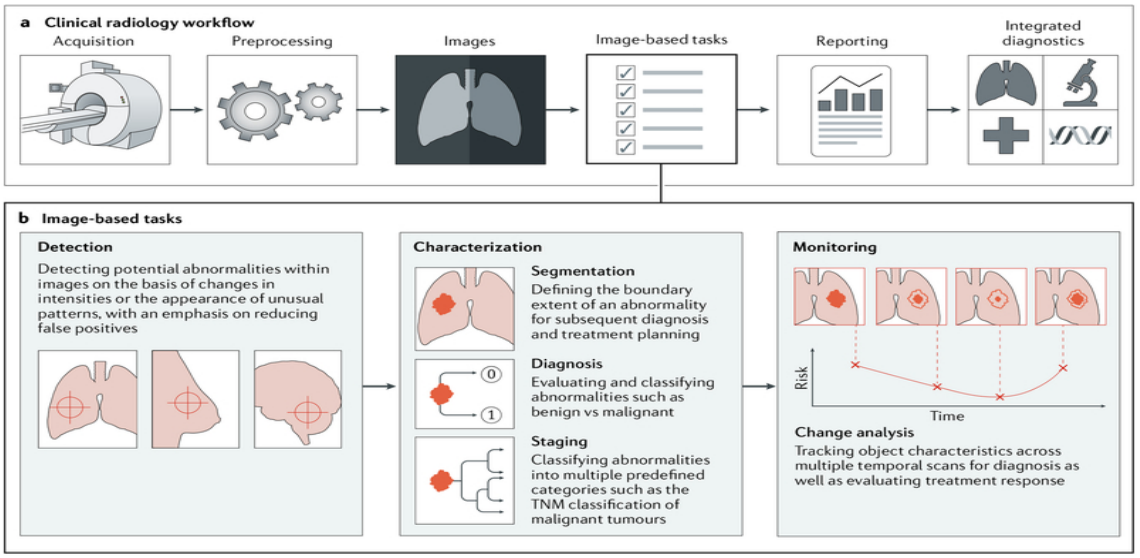

Fig. 7.2 Image showing workflow and case handling in hospitals after introducing AI systems.

Artificial intelligence impacts areas within oncology imaging.

This schematic outlines the various tasks within radiology where artificial intelligence (AI) implementation is likely to have a large impact. a | The workflow comprises the following steps: preprocessing of images after acquisition, image-based clinical tasks (which usually involve the quantification of features either using engineered features with traditional machine learning or deep learning), reporting results through the generation of textual radiology reports and, finally, the integration of patient information from multiple data sources. b | AI is expected to impact image-based

clinical tasks, including the detection of abnormalities; the characterization of objects in images using segmentation, diagnosis and staging; and the monitoring of objects for diagnosis and assessment of treatment response. TNM, tumour–node–metastasis.

There are two classes of AI methods that are in wide use today. The first uses handcrafted engineered features that are defined in terms of mathematical equations (such as tumour texture) and can thus be quantified using computer programs. These features are used as inputs to state-of-the-art machine learning models that are trained to classify patients in ways that can support clinical decision making. Although such features are perceived to be discriminative, they rely on expert definition and hence do not necessarily represent the most optimal feature quantification approach for the discrimination task at hand. Moreover, predefined features are often unable to adapt to variations in imaging modalities, such as computed tomography (CT), positron emission tomography (PET) and magnetic resonance imaging (MRI), and their associated signal-to-noise characteristics.

The second method, deep learning, has gained considerable attention in recent years. Deep learning algorithms can automatically learn feature representations from data without the need for prior definition by human experts. This data-driven approach allows for more abstract feature definitions, making it more informative and generalizable. Deep learning can thus automatically quantify phenotypic characteristics of human tissues, promising substantial improvements in diagnosis and clinical care. Deep learning has the added benefit of reducing the need for manual preprocessing steps. For example, to extract predefined features, accurate segmentation of diseased

tissues by experts is often needed. Because deep learning is data driven (Box 1), with enough example data, it can automatically identify diseased tissues and hence avoid the need for expert-defined segmentations. Given its ability to learn complex data representations, deep learning is also often robust against undesired variation, such as the inter-reader variability, and can hence be applied to a large variety of clinical conditions and parameters. In many ways, deep learning can mirror what trained radiologists do, that is, identify image parameters but also weigh up the importance of these parameters on the basis of other factors to arrive at a clinical decision.

Given the growing number of applications of deep learning in medical imaging, several efforts have compared deep learning methods with their predefined feature-based counterparts and have reported substantial performance improvements with deep learning. Studies have also shown that deep learning technologies are on par with radiologists' performance for both detection and segmentation tasks in ultrasonography and MRI, respectively. For the classification tasks of lymph node metastasis in PET-CT, deep learning had higher sensitivities but lower specificities than radiologists. As these methods are iteratively refined and tailored for specific applications, a better command of the sensitivity:specificity trade-off is expected. Deep learning can also enable faster development times, as it depends solely on curated data and the corresponding metadata rather than domain expertise. On the other hand, traditional predefined feature systems have shown plateauing performance over recent years and hence do not generally meet the stringent requirements for clinical utility. As a result, only a few have been translated into the clinic. It is expected that

high-performance deep learning methods will surpass the threshold for clinical utility in the near future and can therefore be expeditiously translated into the clinic.

8.AI CHALLENGES IN MEDICAL IMAGING

We are currently witnessing a major paradigm shift in the design principles of many computer-based tools used in the clinic. There is great debate about the speed with which newer deep learning methods will be implemented in clinical radiology practice, with speculations for the time needed to fully automate clinical tasks ranging from a few years to decades. The development of deep learning-based automated solutions will begin with tackling the most common clinical problems where sufficient data are available. These problems could involve cases where human expertise is in high demand or data are far too complex for human readers; examples of these include the reading of lung screening CTs, mammograms and images from virtual colonoscopy. A second wave of efforts is likely to address more complex problems such as multiparametric MRI. A common trait among current AI tools is their inability to address more than one task, as is the case with any narrow intelligence. A comprehensive AI system able to detect multiple abnormalities within the entire human body is yet to be developed.

Data continues to be the most central and crucial constituent for learning AI systems. With one out of four Americans receiving a CT examination and one out of ten receiving an MRI examination annually, millions of medical images are produced each year. Additionally, recent well-implemented advances in US-based digital health

systems — such as the Picture Archiving and Communication System (PACS) — have ensured that medical images are electronically organized in a systematic manner, with parallel efforts in Europe and developing countries. It is clear that large amounts of medical data are indeed available and are stored in such a manner that enables moderate ease in access and retrieval. However, such data are rarely curated, and this represents a major bottleneck in attempting to learn any AI model. Curation can refer to patient cohort selection relevant for a specific AI task but can also refer to segmenting objects within images. Curation ensures that training data adheres to a defined set of quality criteria and is clear of compromising artefacts. It can also help avoid unwanted variance in data owing to differences in data-acquisition standards and imaging protocols, especially across institutions, such as the time between contrast agent administration and actual imaging. An example of data curation within oncology could include assembling a cohort of patients with specific stages of disease and tumour histology grades. Although photographic images can be labelled by nonexperts, using, for instance, crowdsourcing approaches, medical images do require domain knowledge. Hence, it is imperative that such curation is performed by a trained reader to ensure credibility — making the process expensive. It is also very time consuming, although utilizing recent deep learning algorithms promises to reduce annotation time substantially: meticulous slice-by-slice segmentation can potentially be substituted by single seed points within the object, from which full segmentations could be automatically generated. The amount of data requiring curation is another limiting factor and is highly dependent on the AI approach — with deep learning methods being more prone to overfitting and hence often requiring more data.

The suboptimal performance of many automated and semi-automated segmentation algorithms has hindered their utility in curating data, as human readers are almost always needed to verify accuracy. More complications arise with rare diseases, where automated labelling algorithms are non-existent. The situation is exacerbated when only a limited number of human readers have previous exposure and are capable of verifying these uncommon diseases. One solution that enables automated data curation is unsupervised learning. Recent advances in unsupervised learning, including generative adversarial networks and variational autoencoders among others, show great promise, as discriminative features are learned without explicit labelling. Recent studies have explored unsupervised domain adaptation using adversarial networks to segment brain MRI, leading to a generalizability and accuracy close to those of supervised learning methods. Others employ sparse autoencoders to segment breast density and score mammographic texture in an unsupervised manner. Self-supervised learning efforts have also utilized spatial context information as supervision for recognizing body parts in CT and MRI volumes through the use of paired CNNs. Nevertheless, public repositories such as The Cancer Imaging Archive (TCIA) offer unparalleled open-access to labelled medical imaging data, allowing immediate AI model prototyping and thus eliminating lengthy data curation steps.

Albeit intuitively leading to higher states of intelligence, the recent paradigm shift from programs based on well-defined rules to others that learn directly from data has brought certain unforeseen concerns to the spotlight. A strong theoretical understanding of deep learning is yet to be established despite the reported successes across many fields — explaining why deep learning layers that lie between inputs and

outputs are labelled as ‘hidden layers’ (Box 1). Identifying specific features of an image that contribute to a predicted outcome is highly hypothetical, causing a lack of understanding of how certain conclusions are drawn by deep learning. This lack of transparency makes it difficult to predict failures, isolate the logic for a specific conclusion or troubleshoot inabilities to generalize to different imaging hardware, scanning protocols and patient populations. Not surprisingly, many uninterpretable AI systems with applications in radiology have been dubbed ‘black-box medicine’ .

From a regulatory perspective, discussions are underway regarding the legal right of regulatory entities to interrogate AI frameworks on the mathematical reasoning for an outcome. While such questioning is possible with explicitly programmed mathematical models, new AI methods such as deep learning have opaque inner workings, as mentioned above. Sifting through hundreds of thousands of nodes in a neural network, and their respective associated connections, to make sense of their stimulation sequence is unattainable. An increased network depth and node count brings more complex decision making together with a much more challenging system to take apart and explore. On the other hand, we find that many safe and effective US Food and Drug Administration (FDA)-approved drugs have unknown mechanisms of action. From that perspective and despite the degree of uncertainty surrounding many AI algorithms, the FDA has already approved high-performance software solutions, though they are known to have somewhat obscure working mechanisms. Regulatory bodies, such as the FDA, have been regulating CADe and CADx systems that rely on machine learning and pattern-recognition techniques since the earliest days of computing. However, it is the shift to deep learning that now poses new regulatory challenges and

requires new guidance for submissions seeking approval. Even after going to market, deep learning methods evolve over time as more data are processed and learned from. Thus, it is crucial to understand the implications of such lifelong learning in these adaptive systems. Periodic testing over specific time intervals could potentially ensure that learning and its associated prediction performance are following forecasted projections. Additionally, such benchmarking tests need to adapt to AI specifics such as the sensitivity of prediction probabilities in CNNs.

Other ethical issues may arise from the use of patient data to train these AI systems. Data are hosted within networks of medical institutions, often lacking secure connections to state-of-the-art AI systems hosted elsewhere. More recently, Health Insurance Portability and Accountability Act (HIPAA)-compliant storage systems have paved the way for more stringent privacy preservation. Studies have explored systems that enable multiple entities to jointly train AI models without sharing their input data sets — sharing only the trained model. Other efforts use a decentralized 'federated' learning approach. During training, data remains local, while a shared model is learned by combining local updates. Inference is then performed locally on live copies of the shared model, eliminating data sharing and privacy concerns. 'Cryptonets' are deep learning networks trained on encrypted data, and they even make encrypted predictions that can be decrypted only by the owner of a decryption key — thus ensuring complete confidentiality throughout the entire process. All these solutions, albeit still in early developmental stages, promise to create a sustainable 'data to AI' ecosystem — without undermining privacy and HIPAA compliance.

9.FUTURE PERSPECTIVES

From the early days of X-ray imaging in the 1890s to more recent advances in CT, MRI and PET scanning, medical imaging continues to be a pillar of medical treatment. Current advances in imaging hardware — in terms of quality, sensitivity and resolution — enable the discrimination of minute differences in tissue densities. Such differences are, in some cases, difficult to recognize by a trained eye and even by some traditional AI methods used in the clinic. These methods are thus not fully on par with the sophistication of imaging instruments, yet they serve as another motivation to pursue this paradigm shift towards more powerful AI tools. Moreover, and in contrast to traditional methods based on predefined features, we find that deep learning algorithms scale with data, that is, as more data are generated every day and with ongoing research efforts, we expect to see relative improvements in performance. All these advances promise an increased accuracy and reduction in the number of routine tasks that exhaust time and effort.

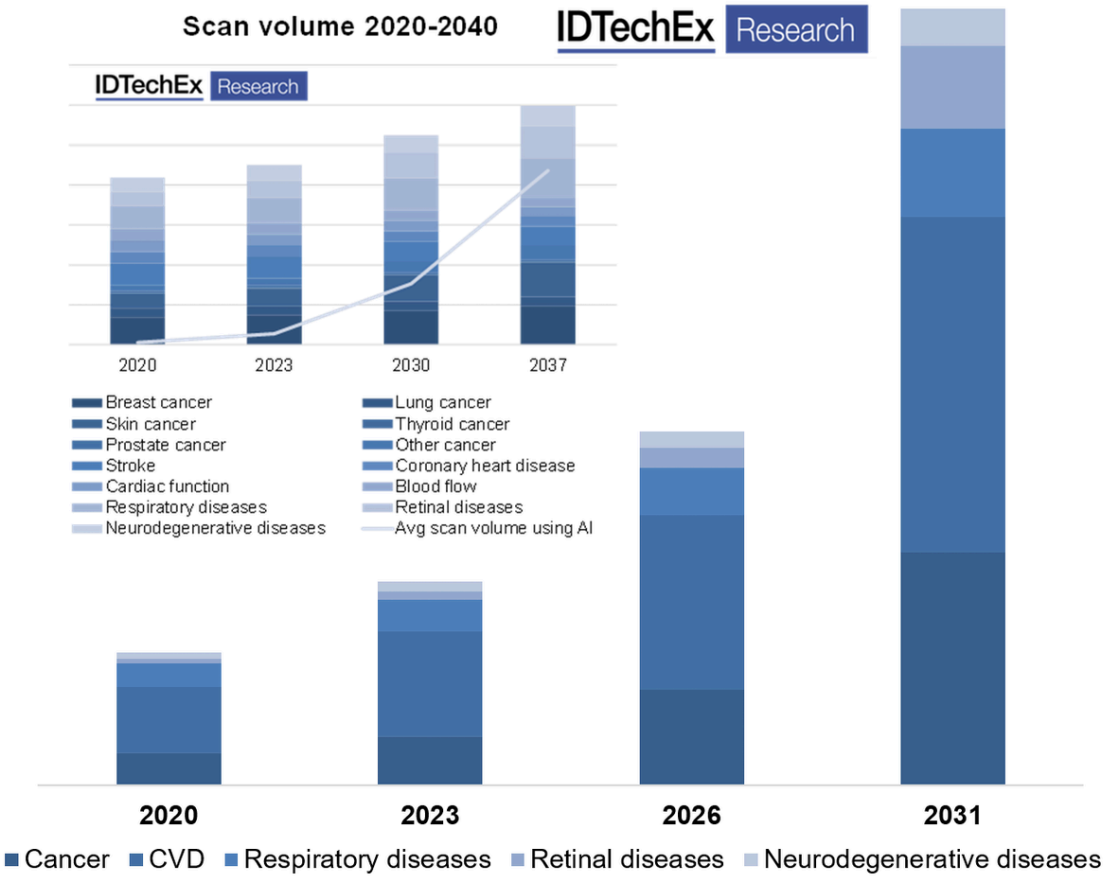

Fig. 9.1 Future funding on Medical imaging in relation with AI

Aligning research methodologies is crucial in accurately assessing the impact of AI on patient outcome. In addition to the undeniable importance of reproducibility and generalizability, utilizing agreed-upon benchmarking data sets, performance metrics, standard imaging protocols and reporting formats will level the experimentation field and enable unbiased indicators. It is also important to note that AI is unlike human intelligence in many ways; excelling in one task does not necessarily imply excellence in others. Therefore, the promise of up-and-coming AI methods should not be overstated. Almost all state-of-the-art advances in the field of AI fall under the narrow AI category, where AI is trained for one task and one task only — with only a handful

exceeding human intelligence. While such advances excel in interpreting sensory perceptual information in a bottom-up fashion, they lack higher-level, top-down knowledge of contexts and fail to make associations the way a human brain does. Thus, it is evident that the field is still in its infancy, and overhyped excitement surrounding it should be replaced with rational thinking and mindful planning. It is also evident that AI is unlikely to replace radiologists within the near or even distant future. The roles of radiologists will expand as they become more connected to technology and have access to better tools. They are also likely to emerge as critical elements in the AI training process, contributing knowledge and overseeing efficacy. As different forms of AI exceed human performance, we expect it to evolve into a valuable educational resource. Human operators will not only oversee outcomes but also seek to interpret the reasoning behind them — as a means of validation and as a way to potentially discover hidden information that might have been overlooked .

In contrast to traditional AI algorithms locked within proprietary commercial packages, we find that the most popular deep learning software platforms available today are open-source. Raw data are almost always downsampled and optimized for human viewers. This simplification and loss of information are both avoidable when the analyses are run by machines but are associated with caveats including reduced interpretability and impeded human validation. As more data are generated, more signals are available for training. However, more noise is also present. We expect the process of discerning signal from noise to become more challenging over time. With difficulties in curating and labelling data, we foresee a major push towards unsupervised learning techniques to fully utilize the vast archives of unlabelled data.

10.CONCLUSION

Open questions include the ambiguity of who controls AI and is ultimately responsible for its actions, the nature of the interface between AI and health care and whether implementation of a regulatory policy too soon will cripple AI application efforts. Enabling interoperability among the multitude of AI applications that are currently scattered across health care will result in a network of powerful tools. This AI web will function at not only the inference level but also the lifelong training level. We join the many calls that advocate for creating an interconnected network of de-identified patient data from across the world. Utilizing such data to train AI on a massive scale will enable a robust AI that is generalizable across different patient demographics, geographic regions, diseases and standards of care. Only then will we see a socially responsible AI benefiting the many and not the few. Deep learning has already shown comparable performance to humans in recognition and computer vision tasks. These technological changes make it reasonable to think that there might be some major changes in clinical practices. When we consider the use of AI in medical imaging, we anticipate this technological innovation to serve as a collaborative medium in decreasing the burden and distraction from many repetitive and humdrum tasks, rather than replacing physicians. The use of deep learning and AI in radiology is currently in the stages of infancy. One of the key factors for the development and its proper clinical adoption in medicine would be a good mutual understanding of the AI technology, and the most appropriate form of clinical practice and workflow by both clinicians and computer scientists/engineers.

REFERENCES

1. Ahmed Hosny, Chintan Parmar, John Quackenbush, Lawrence H. Schwartz, and Hugo J. W. L. Aerts, Artificial intelligence in radiology. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6268174/
2. Emre Pakdemirli, Artificial intelligence in radiology: friend or foe? Where are we now and where are we heading? https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6385326/
3. James H Thrall, Xiang Li, Quanzheng Li, Cinthia Cruz, Synho Do, Keith Dreyer, James Brink, Artificial Intelligence and Machine Learning in Radiology: Opportunities, Challenges, Pitfalls, and Criteria for Success. https://pubmed.ncbi.nlm.nih.gov/29402533/
4. Pranav Rajpurkar, Jeremy Irvin, Robyn L Ball, Kaylie Zhu, Brandon Yang, Hershel Mehta, Tony Duan, Daisy Ding, Aarti Bagul, Curtis P Langlotz, Bhavik N Patel, Kristen W Yeom, Katie Shpanskaya, Francis G Blankenber, Jayne Seekins, Timothy J Amrhein, David A Mong, Safwan S Halabi, Evan J Zucker, Andrew Y Ng, Matthew P Lungren, Deep learning for chest radiograph diagnosis: A retrospective comparison of the CheXNeXt algorithm to practicing radiologists.

   https://pubmed.ncbi.nlm.nih.gov/30457988/
5. Sara Reardon, Rise of Robot Radiologists. *Scientificamerican* [online] 2020.